Supplemental material: the measured grid, the reduction stages, the production environment, and the members as dispatched

S1.1 The measured grid, cell by cell

Across the three directories, `IR` covers $N_{\mathrm{ch}} \in \{5, 10, 20, 50, 100, 200, 500, 1000, 2000, 5000, 10000\}$ and all twelve levels; NR covers $N_{\mathrm{ch}} \in \{10, 100, 1000, 10000\}$ and R covers those and additionally 5, 20 and 50 at $\widetilde{S} = 0.01$, with the highest levels run only at the larger channel counts and reaching $\widetilde{S} = 10^6$ at $N_{\mathrm{ch}} = 10^4$; INR, MR and VR cover the same 30 combinations of channel number and noise as the least-squares arm, which runs on $N_{\mathrm{ch}} \in \{10, 100, 1000, 10000\}$ with $\widetilde{S}$ from 0.01 up to 10^3, 10^4, 10^5 and 10^6 respectively. Those 30 combinations, times the seven intervals, are the 210 cells every one of the eight members carries; IR carries 560.

S1.2 The five reduction stages and what each writes

Each grid cell is reduced through five stages: the per-group MLE cloud; the pooled joint fit θ_{pool}; the score at both anchors, whose across-replicate covariance is J and whose per-interval Gaussian terms sum to the anchor $\bar{G}$; the diagnostic battery at each anchor; and the empirical-versus-theoretical distortion capstone, anchored at θ_{pool}. Each cell writes five comma-separated-value families (the estimate cloud, the pooled fit, the two anchor batteries, and the capstone). The scripts number these stages one, two, four, five and six; the gap at three is the numerical-Fisher stage, which the Gaussian anchor removes and which the numbering keeps visible. On the grid-production lane the cloud stage and the capstone are dropped, because no map reads them there; the cells that need a cloud are produced by the separate script that keeps that stage.

S1.3 The production environment

The production batteries ran under SLURM at 32 CPUs per task and 48 GB of memory (96 GB for the parameter-recovery lane), with the linear-algebra back end pinned to a single thread so that OpenMP parallelises the per-simulation and bootstrap loops. The environment variable `MACRODR_AXIS_SERIAL=1` is load-bearing: it serialises the internal grid-combination loop, without which memory grows with the number of concurrent combinations and the jobs exhaust memory.

S1.4 The optimiser schedule and its tolerances

The script sets the damping schedule (initial damping 10, growth factor 3, maximum damping 10^{10}) and three limits: at most 100 iterations, a relative gradient tolerance of 10^{-6} and a value tolerance of 10^{-10}. The Newton-decrement criterion that stops most fits is described in Methods.

S1.5 The differenced-Fisher check of the Gaussian anchor

The differenced Fisher information $\mathbf{F}$ is a central difference of the analytic score, with per-coordinate step $h_i = h_{\mathrm{rel}} \max(|\theta_i|, 1)$ at $h_{\mathrm{rel}} = 10^{-5}$, symmetrised as $(\mathbf{F} + \mathbf{F}^{\mathsf{T}})/2$ and averaged over recordings before any decomposition. Because the score itself is obtained by automatic differentiation, this is a single numerical derivative rather than two. The check is the congruence $\mathbf{C} = \mathbf{G}^{-1/2}\,\mathbf{F}\,\mathbf{G}^{-1/2}$ with $\mathbf{G}$ the analytic Gaussian Fisher information that anchors every distortion in the body: $\mathbf{C}$ at the identity means the cheap analytic form carries the member's true curvature, and an entry of the table is the geometric mean of $\mathbf{C}$'s eigenvalues, the median over the seven acquisition intervals, at $\widetilde{S} = 0.01$ and $n_{\mathrm{sim}} = 10^4$ exact simulations per cell, under one explicit injected seed for the whole sweep, which also pairs the ensemble across members within a cell.

member	$N_{\mathrm{ch}} = 10$	10^2	10^3	10^4
LSE	0.71	0.78	$0.78^{\dagger}$	$0.55^{\dagger}$
ILSE	0.71	0.78	0.78	0.71
NR	$0.91^{\dagger}$	$0.99^{\dagger}$	$1.16^{\dagger}$	$1.36^{\dagger}$
INR	1.00	1.00	1.00	1.00
R	$0.91^{\dagger}$	0.87	$0.94^{\dagger}$	$0.94^{\dagger}$
MR	$0.90^{\dagger}$	$0.87^{\dagger}$	0.91	0.91
VR	1.02	0.94	0.95	0.94
IR	0.99	1.00	1.00	1.00

†one or more of the seven intervals carries an indefinite differenced Fisher and is excluded from that median; for `NR` at $N_{\mathrm{ch}} = 10^4$ the three coarsest intervals are indefinite and the retained maximum is 1.97 at $\widetilde{\Delta} = 0.1$. The two least-squares rows are 4×4 over the rates and the amplitude pair the fit estimates; every other row is 6×6 (Methods).

Three readings. For `INR` and `IR` the two constructions coincide within one per cent at every channel count, so for the calibrated member and its non-recursive sibling the anchor question closes. For `NR` the two part company as channels are added: re-anchoring `NR`'s distortions on the differenced Fisher divides them by about the median above, so its largest reported factors at 10^4 channels narrow by roughly a third without being overturned, and at the coarsest intervals the differenced Fisher is not positive at all, which is the same ill-conditioned corner the body figure greys. For both least-squares arms the entries sit near 0.7–0.8, so re-anchoring would enlarge their distortions by about 1.3: the least-squares verdict in the body is conservative under the anchor. At ten channels the differenced construction itself is noisy (single-interval entries for `IR` range 0.50 to 1.38 around a median of 0.99), which is finite-difference conditioning at small information, not structure.

The earlier battery in data directory `433ed13`, superseded and quoted nowhere in the body, gave the same picture on the four members it carried (medians 1.011 for `R`, 1.020 for `MR`, 1.000 for `IR`; `NR` rising 1.26 to 1.60 across the two largest channel counts at this noise); it lacked both least-squares arms and `INR` entirely, which is what the run above repairs.

S1.6 The members as dispatched: flags, data keys and cells

Least squares runs non-recursive, and the variance flag is accepted and ignored on that branch. The averaging flag still applies to it, and it is what separates the two least-squares arms: at 0 the fit predicts the deterministic mean current at one instant, the middle of the acquisition window (`LSE`, data key `nonlinearsqr_g`), and at 1 it predicts that same mean averaged over the window

(`ILSE`, data key `nonlinearsqr`). Each arm shares its mean model exactly with the macroscopic member carrying the same averaging flag, `LSE` with `NR` and `ILSE` with `INR`, and differs from it only in the predictive variance: on the recording of Figure 1 of the manuscript the predicted means of `LSE` and `NR` agree to 9×10^{-16} interval by interval, and those of `ILSE` and `INR` to 2×10^{-16}, while the two least-squares arms differ from each other by 5×10^{-2} on the same intervals. Both arms sit off the lattice of endpoint and recursion flags that organises the macroscopic members; the `I` prefix records the window setting and does not place `ILSE` on that lattice.

Both least-squares arms are dispatched over the design grid, and over the same cells: the thirty combinations of channel number and noise level swept at ten thousand simulations are run with `ILSE` and with `LSE` alike, from the same script that produces the ensemble the other members score. Four further cells exist for `ILSE` at a thousand simulations, superseded by the ten-thousand-simulation run of the same cells, and two at $N_{\mathrm{ch}} = 5$, below the range the paper reports.

The macroscopic members are one filter run under flags that set how the interval-averaged conductance is conditioned. The *recursive* flag sets whether the occupancy covariance is propagated between intervals. The *averaging* flag av $\in \{0, 1, 2\}$ counts the number of interval endpoints the hidden state is conditioned on: 0 an instant, 1 the state at the window's start, 2 the states at both of its ends. A third flag, the *variance form*, distinguishes the two members that condition on the interval start: `MR` predicts the interval variance with the total per-start-state conductance variance, while `VR` removes the between-endpoint part, keeping `MR`'s mean and gain.

Member	Data key	Recursive	av	Variance	Conditioning of the conductance
LSE	`nonlinearsqr_g`	no	0	n/a	none; the deterministic mean current at one instant, the window's middle
ILSE	`nonlinearsqr`	no	1	n/a	none; the deterministic mean current averaged over the interval
NR	`macro_NR`	no	0	n/a	instantaneous; the acquisition averaging is ignored
INR	`macro_INR`	no	1	total	interval mean, without recursion; the endpoints are not separable
R	`macro_R`	yes	0	n/a	instantaneous; the acquisition averaging is ignored
MR	`macro_MR`	yes	1	total	interval mean given the start state
VR	`macro_VR`	yes	1	residual	MR's mean and gain, with the between-endpoint variance removed
IR	`macro_IR`	yes	2	residual	interval mean given both boundary states; the endpoints enter the gain

The members as flags. Both least-squares arms take *family approximation* = 2 and every macroscopic member takes 0. Reading the endpoint flag as a ladder gives no endpoints (`NR`, `R`), one endpoint (`MR`, `VR`), then both boundary states (`IR`); `INR` sits outside that ordering, because without a recursive update the boundary state is unobservable and conditioning on the start state and conditioning on both ends describe the same method. The variance column reports the form in force rather than the dispatched flag: `IR` is dispatched with the total flag, but with both endpoints conditioned the residual form is used regardless of it, because the between-endpoint spread is already carried by the boundary term. Even so, taken from the same state `MR` and `IR` assign the interval average the same total variance, by two decompositions that rearrange into each other (Appendix 2 of the manuscript), so what separates them is the gain; because that variance divides the gain, the states they carry part company at the first gated update and the variances they report along a recording part with them. `VR` keeps `MR`'s mean and boundary-free gain and replaces the total per-start-state conductance variance with the residual one, which is not the same thing as carrying `IR`'s variance. The least-squares arms and the macroscopic members are written by different producers, so their output files carry different prefixes, `figure_3_LSE_` and `figure_3_G_`. **The identification of `INR` with the published control.** `INR` is the member published as MacroINR by Moffatt and Pierdominici-Sottile (2025), where it served as the control against which MacroIR, the member called `IR` in the manuscript, was compared. The deposited run metadata of that study confirm the identification: its two arms differ in the recursion flag alone, every other flag being constant across the 414 deposited run records. The deposited endpoint flag coincides with this study's `INR` exactly, because without a recursive update conditioning on one end and conditioning on both describe the same likelihood; evaluated on the reference binary, the two settings return byte-identical output.

S1.7 Symbols against implementation names, and the fixed build flags

The equations of Appendix 1 of the manuscript are written with the objects the implementation forms, and the correspondence is one to one. $\circ$ is the entrywise product and $\mathbf{1}$ the vector of ones; $\overline{m}_{i\to j} = \mathbb{E}[\Delta^{-2} A^2_{0\to\Delta} \mid X_0 = i, X_\Delta = j]$ is the pair-conditioned second moment, so that $\bar{v}_{i\to j} = \overline{m}_{i\to j} - \bar{\gamma}^2_{i\to j}$.

symbol	in the implementation	what it is
$P_{i\to j}(\Delta)$	`P`	transition over the window
$\bar{\gamma}_{i\to j}$	`gmean_ij`	conductance averaged over the window, given the pair
$G_{i\to j} = \bar{\gamma}_{i\to j} P_{i\to j}$	`gtotal_ij`	the same, weighted, so that it can be summed
$\overline{m}_{i\to j}$	`gsqr_ij`	its second moment, given the pair
$(\bar{\gamma}_0)_i$	`gmean_i`	= `gtotal_ij` $\cdot\,\mathbf{1}$
$\overline{m}_{i\to\cdot}$	`gsqr_i`	= `gtotal_sqr_ij` $\cdot\,\mathbf{1}$
$H_{i\to j}$, collapsed	(`gtotal_ij` $\circ$ `gmean_ij`)$\cdot\mathbf{1}$	the square collapsed, not the collapse squared
$\bar{v}_{i\to j}$	`gvar_ij`	present in the window object, unused by the recursion
$\boldsymbol{\mu}_0$, $\boldsymbol{\Sigma}_0$	`P_mean`, `P_Cov`	the state where the window opens
$\boldsymbol{\Sigma}_0 - \operatorname{diag}\boldsymbol{\mu}_0$	`SmD`	the prior with its single-channel part removed
$\boldsymbol{\gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma}$ and its tilde	`gSg`	what the state contributes to the variance
$\boldsymbol{\mu}^\top\mathbf{w}$	`ms`	the state-dependent noise
$\mathbf{g}_{0\to\Delta}$	`gS`	state against current
$\sigma^2_{0\to\Delta}$	`y_var`	the predictive variance

Throughout the reported runs the remaining build flags were held fixed: *taylor variance correction* off, *micro approximation* off, and the *variance approximation* on. The V in `VR` names the *variance form* flag, set to 1, and has no connection to the *taylor variance correction* flag, which is off in every run reported here; the Taylor-corrected variants MRT and IRT are not part of the study. Because the Taylor variance correction is off, the recursive one-endpoint member that runs is MacroMR and not MacroMRT, which is what a reader matching deposited filenames needs to know.

S1.8 Design rationale behind the two numerical safeguards

Why ε is keyed to the conditioning, and where the prior comes from. Both choices in the pseudo-count are deliberate. Keying ε to $\kappa_F(\mathbf{V})$ ties the regularizer to the conditioning of the decomposition that produced the noise rather than to any model constant. Taking the prior means from the conductance vector rather than from the marginals of $\mathbf{F}_1$ keeps it clean when $\mathbf{V}$ is ill conditioned, which is exactly when those marginals carry the same reconstruction error the pseudo-count exists to absorb and when the prior is most needed.

Why the step is damped rather than the state corrected. The earlier form of this code took the undamped step and truncated the occupancy entrywise at zero before renormalising, which is not differentiable where it acts and so carries, one level up, the defect the smooth trust coefficient exists to avoid. With the step damped instead, no operation produces a negative entry, and the renormalisation that follows validates rather than repairs, failing on a real one.

The two trusts, and the size of the smoothing bias. The covariance is treated separately and takes its full step, since the rank-one down-date of the posterior covariance is positive semidefinite by construction; the two trusts are not tied to a common factor. In the smooth coefficient, $c = 0.9$ is a safety factor applied when the constraint binds, leaving a ten per cent margin against the boundary, and $\epsilon = 10^{-4}$ sets the width of the smoothing band, so the residual bias of α below the hard minimum is at most $\epsilon/2$, three orders of magnitude smaller than that margin. A merely continuous coefficient is not enough and a discontinuous one leaves a delta in the score.